

Unit 2: First Order Linear Differential Equations — Cheat Sheet

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Condensed Cheat Sheet of Core Concepts and Master Formulas

Unit 2 at a Glance: Core Sub-Topics

This unit assembles the core solving toolkit of first order theory. Each sub-topic below builds on the last, moving from raw integration techniques toward genuine physical models.

- **Separable Equations** — splitting a rate law into a pure x side and a pure y side, then integrating each independently.
- **First-Order Linear Equations and Integrating Factors** — the standard form $y' + P(x)y = Q(x)$ and the factor that collapses the left side into a single derivative.
- **Initial Value Problems** — using a known data point to pin the arbitrary constant and select one curve from the solution family.
- **Homogeneous Equations and Substitution** — the substitution $y = vx$ that reduces a degree-matched equation to a separable one.
- **Applications: Growth, Decay, and Newton's Law of Cooling** — the exponential models that these methods unlock.

Part I — Condensed Cheat Sheet

2.1 Separable Equations

A first order equation is *separable* when its right side factors into a function of x alone times a function of y alone — that single structural test is the whole game. Once it factors, you divide and multiply until every y rides with dy and every x rides with dx , then integrate both sides.

Standard Form and Method: Separable

A separable equation and its solution scheme:

$$\frac{dy}{dx} = g(x) h(y) \implies \int \frac{1}{h(y)} dy = \int g(x) dx + C.$$

One constant C suffices, since the two constants of integration merge into a single arbitrary constant.

Pitfall. Dividing by $h(y)$ silently assumes $h(y) \neq 0$. Any constant value of y that makes $h(y) = 0$ is an equilibrium solution that this step can quietly discard — always check it separately.

Worked Example 2.1: Solve $\frac{dy}{dx} = 6x^2y$ **Step 1.** Separate the variables:

$$\frac{1}{y} dy = 6x^2 dx.$$

Step 2. Integrate both sides:

$$\ln |y| = 2x^3 + C.$$

Step 3. Exponentiate to isolate y :

$$y = e^{2x^3+C} = C e^{2x^3}.$$

The dropped solution $y = 0$ is recovered as the case $C = 0$.**2.2 First-Order Linear Equations and Integrating Factors**

A first order equation is *linear* when y and y' each appear only to the first power and never multiply one another. Before reading off anything, you must normalize so the derivative carries coefficient one — only then is $P(x)$ truly the coefficient of y .

Standard Form and Integrating Factor

Standard form, integrating factor, and solution:

$$\frac{dy}{dx} + P(x)y = Q(x), \quad \mu(x) = e^{\int P(x) dx},$$

$$\frac{d}{dx}[\mu(x)y] = \mu(x)Q(x) \implies y = \frac{1}{\mu(x)} \left(\int \mu(x)Q(x) dx + C \right).$$

The magic of $\mu(x)$ is that multiplying through turns the entire left side into the exact product-rule derivative $\frac{d}{dx}[\mu y]$ — that is the single fact the whole method rests on.

Worked Example 2.2: Solve $\frac{dy}{dx} + y = e^x$ **Step 1.** Read off $P(x) = 1$, so the integrating factor is

$$\mu = e^{\int 1 dx} = e^x.$$

Step 2. Multiply through; the left side collapses to a derivative:

$$\frac{d}{dx}[e^x y] = e^x \cdot e^x = e^{2x}.$$

Step 3. Integrate both sides:

$$e^x y = \frac{1}{2} e^{2x} + C.$$

Step 4. Divide by e^x :

$$y = \frac{1}{2} e^x + C e^{-x}.$$

2.3 Initial Value Problems

The general solution of a first order equation is a whole family of curves indexed by one arbitrary constant. An *initial condition* $y(x_0) = y_0$ selects exactly one member — it fixes the constant to the value that forces the curve through the given point.

General versus Particular Solution

A first order equation needs exactly **one** initial condition. The general solution carries the free constant; the particular solution has it determined:

$$y = F(x) + C \xrightarrow{y(x_0)=y_0} C = y_0 - F(x_0).$$

Worked Example 2.3: Solve $\frac{dy}{dx} = 3x^2$, $y(1) = 5$

Step 1. Integrate for the general solution:

$$y = x^3 + C.$$

Step 2. Apply $y(1) = 5$:

$$5 = (1)^3 + C \implies C = 4.$$

Step 3. State the particular solution:

$$y = x^3 + 4.$$

2.4 Homogeneous First-Order Equations and Substitution

A function $f(x, y)$ is *homogeneous of degree n* when scaling both inputs pulls out a clean power of the scale factor: $f(tx, ty) = t^n f(x, y)$. An equation $M dx + N dy = 0$ is homogeneous when M and N share the same degree — and that shared degree is precisely what lets the ratio substitution work.

The Substitution $y = vx$

For a homogeneous equation, set

$$y = vx, \quad \frac{dy}{dx} = v + x \frac{dv}{dx},$$

which always reduces the equation to a *separable* one in v and x . After solving for v , back-substitute $v = \frac{y}{x}$.

Worked Example 2.4: Solve $\frac{dy}{dx} = \frac{x+2y}{x}$

Step 1. Rewrite the right side and substitute $y = vx$:

$$\frac{dy}{dx} = 1 + 2\frac{y}{x} = 1 + 2v, \quad v + x \frac{dv}{dx} = 1 + 2v.$$

Step 2. Cancel v and separate:

$$x \frac{dv}{dx} = 1 + v \implies \frac{1}{1+v} dv = \frac{1}{x} dx.$$

Step 3. Integrate and exponentiate:

$$\ln |1+v| = \ln |x| + C \implies 1+v = Cx.$$

Step 4. Back-substitute $v = \frac{y}{x}$ and clear:

$$1 + \frac{y}{x} = Cx \implies y = Cx^2 - x.$$

2.5 Applications: Growth, Decay, and Newton's Law of Cooling

When a rate of change is proportional to the current amount, the solution is exponential — this single idea drives population models, radioactive decay, and cooling alike.

Exponential Models

Growth and decay:

$$\frac{dP}{dt} = kP \implies P(t) = P_0 e^{kt} \quad (k > 0 \text{ grows, } k < 0 \text{ decays}),$$

with doubling time $t = \frac{\ln 2}{k}$. Newton's law of cooling:

$$\frac{dT}{dt} = -k(T - T_s) \implies T(t) = T_s + (T_0 - T_s)e^{-kt},$$

so $T \rightarrow T_s$ as $t \rightarrow \infty$.

Worked Example 2.5: A culture doubles in 3 hours

Step 1. The model $P = P_0 e^{kt}$ doubles when $e^{3k} = 2$:

$$3k = \ln 2 \implies k = \frac{\ln 2}{3}.$$

Step 2. The growth law is therefore

$$P(t) = P_0 e^{(\ln 2/3)t} = P_0 2^{t/3}.$$

Every 3 hours the population multiplies by two, as required.